

HUNTER MATHIAS GILL

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GitHub: [hmgill](https://github.com/hmgill) | HuggingFace: [hmgill](https://huggingface.co/hmgill) | Publications: <https://scholar.google.com/citations?user=geEx8o8AAAAJ&hl=en>

OVERVIEW

AI Engineer & Applied Scientist with 6 years of experience developing multimodal AI systems, LLM agents, and computer vision models for healthcare and regulated environments. Expertise in agentic workflows, MCP integrations, RAG systems, and explainable AI. Proven record of deploying production-ready AI applications using cloud-native infrastructure.

WORK EXPERIENCE

Contractor, Mercor

May 2026 – Present

- Roles: **Generalist, Computer Science Expert**
- Designed quality assessment rubrics for client image generation models, maintaining more than 90% billable utilization over course of projects

Research Fellow, Indiana University Indianapolis

August 2019 – October 2025

- Built REST-based vision loss simulator with FastAPI | <https://vf-simulator-22drbvsetq-uc.a.run.app/api/docs>
- Designed CAMVAS blood vessel segmentation model, pretrained on 1,000 blood vessel images. 99% accuracy, 86% IoU (intersection-over-union) score. Computed 25 vascular descriptors for proprietary chorioallantoic membrane images to perform *C. infortunatum* extract bioactivity profiling | <https://github.com/hmgill/camvas/>
- Trained RetinaRadar model on 20,000 images to assess quality (clarity, artifacts, field-of-view, usability) of standard, widefield, and ultrawidefield retina fundus photos | <https://huggingface.co/spaces/hmgill/retinaradar-demo>
- Developed BigEye by annotating and training on 1,000 retina fundus images, achieving 83% stage-specific accuracy in DR stage detection. Added SHAP analysis for extra model explainability | <https://github.com/Janga-Lab/BigEye>
- Provided technical mentoring and design documentation for a team of 5 Masters students to develop a braces detection model. Trained on multi-source dataset of 4,000 dental panoramic radiographs, with accuracy of 95% and IoU score of 83%

SELECTED PROJECTS

- Designed sandboxed GPT 5.5 agent for assessing chest x-rays (CXR). MCP connections to NVIDIA CXR-Reason and LocateAnything models for narrative writing and finding localization | <https://huggingface.co/spaces/hmgill/CXR-Agent>
- Built sandboxed GPT 5.4 agent for interpreting Optical Coherence Tomography (OCT) scans, with MCP connections for specialist LO-VLM and MIRAGE foundation models | <https://huggingface.co/spaces/hmgill/OCT-Agent>
- Created and deployed Gemini 3 multi-agent system to segment cellular structures with SAM3, with GraphRAG-powered parameters selection. 75% faster per image compared to manual annotation for GinkGo GDPx3 dataset | <https://huggingface.co/spaces/hmgill/Cellemetry>
- Deployed and managed low-latency MCP servers with Modal serverless GPU backends for retinal fundus analysis | cup-disc: <https://fundus-cup-disc-seg.fastmcp.app/mcp>, artery/vein, fovea: <https://fundus-vascx.fastmcp.app/mcp>

SKILLS

Machine Learning, LLMs, Agents

Frameworks (Tensorflow, Pytorch, Huggingface Transformers, sklearn), **LLMs & VLMs** (Gemma, Gemini, Claude Sonnet, GLM, GPT-OSS), **Agent Frameworks** (Google ADK, OpenAI Agents SDK, smolagents), **Vector Databases** (Pinecone, chromadb), **Knowledge Graphs** (Neo4J), **RAG** (langchain), **MCP** (FastMCP, Prefect Horizon), **Prompt Versioning & Observability** (Langfuse, PromptLayer), **Reproducible Workflows** (MLflow), **Data Models** (pydantic), **Traditional ML and Deep Learning** (SVM, CNN, image classification / segmentation, transfer learning, computer vision, RF, LGBM, XGBoost, feature engineering, data cleaning, numpy, pandas)

Programming Languages, Computing Systems, Containerization

Python, R, Typescript, Javascript, Linux, JSON, AJAX, XML, PHP, SQL, MySQL Server, Siteimprove, DAX, APIs, HPC, Slurm, Pyspark, GCP, Vertex AI / Google Enterprise Engine, FastAPI, NextFlow, Dask, Git, GitHub CI/CD, Apptainer, Docker, Modal

Data Processing, Dashboards, Visualizations

ELT, DataBricks, Label Studio, Roboflow, Gradio, Streamlit, Plotly, RShiny, Jupyter, Tableau, Power BI, Figma, BioRender

EDUCATION

Ph.D. in Bioinformatics and Minor in Applied Data Science

October 2025

Luddy School of Informatics, Computing, and Engineering at Indiana University Indianapolis

B.Sc. in Chemistry

May 2019

College of Arts and Sciences at Indiana State University